California Coast Ecological Conservation

Coupling Satellite Observations and Machine Learning to Understand Future Fog Prevalence along the Coast Redwood Range

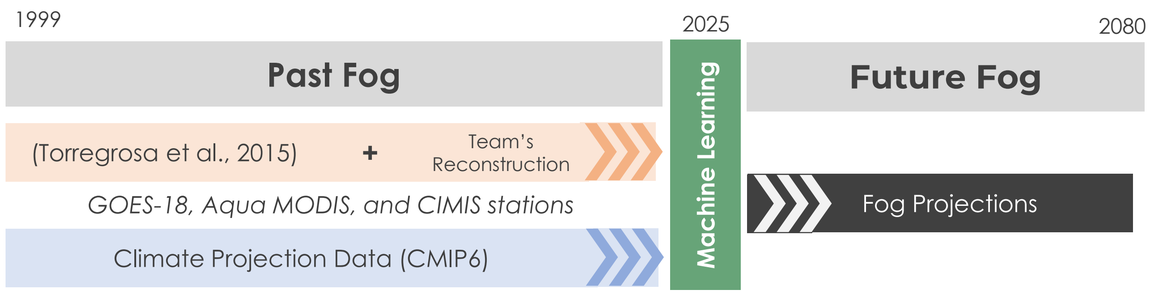
Project Synopsis

The California Coast Ecological Conservation team sought to inform coast redwood conservation by understanding fog prevalence—past, present, and future—across the California coast. Using satellite-based remote sensing and machine learning, the team analyzed past fog frequency (June–September) from 2023–2024, examined sea and land surface temperatures, and created fog projections to the year 2090. These findings show our partner, Save the Redwoods League, current fog patterns and how they may change, helping them prioritize areas for redwood conservation.

Project Partner

- Save the Redwoods League

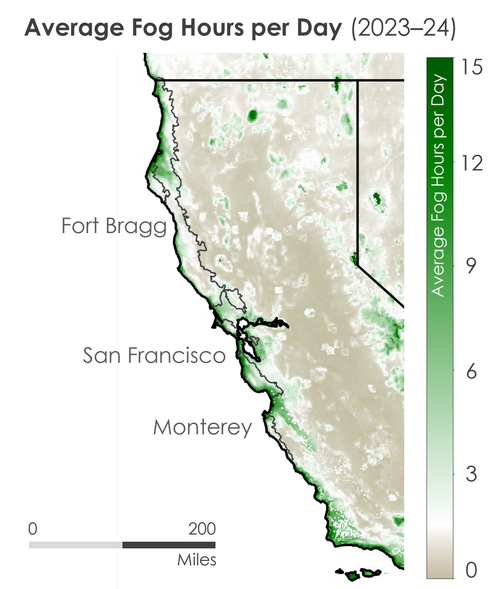
Methods



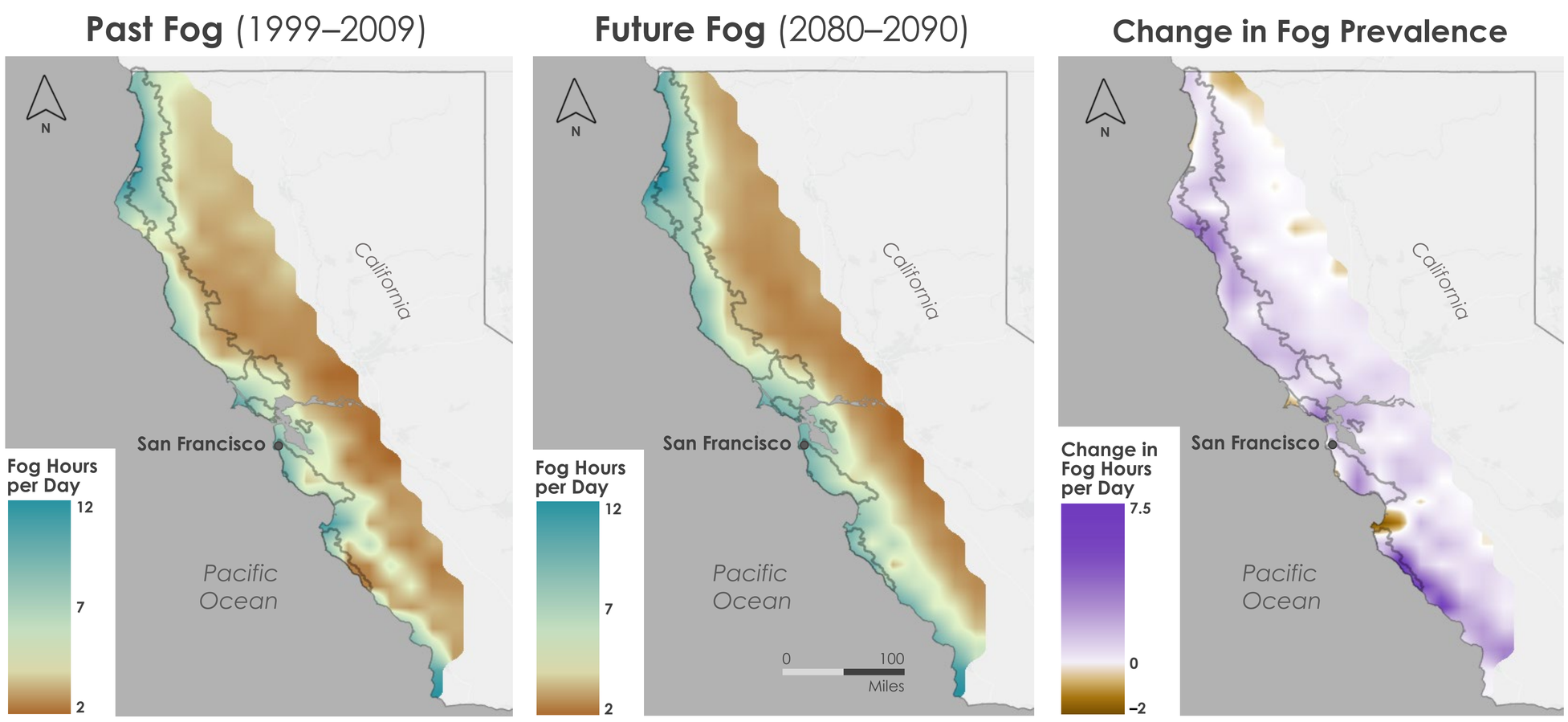
Objectives

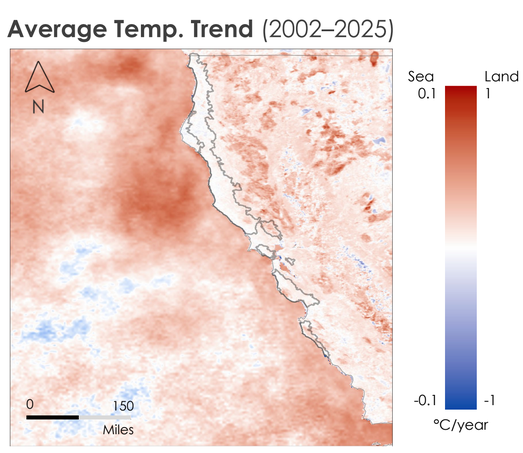
- **Build** a historical fog/low cloud model
- **Understand** important driver variables, like sea and land surface temperature
- **Predict** future fog frequency in 2080–2090 with a random forest regressor (SSP2-8.5)
- **Identify** regions where fog/low cloud will be stable or increasing in the future

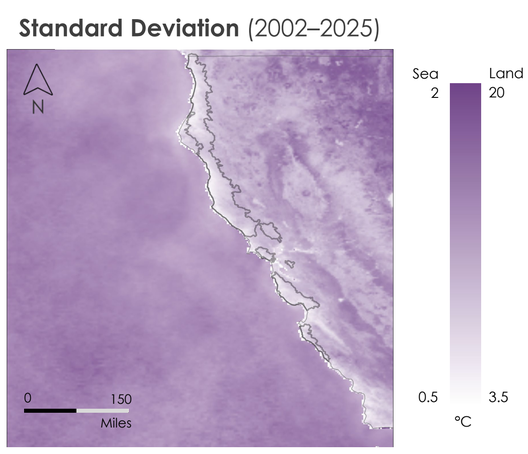
Study Area



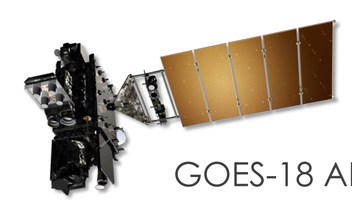
Results







Earth Observations



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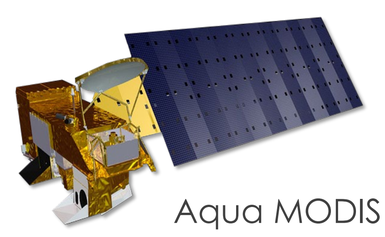

Aqua MODIS

Conclusions

- Fog is increasing across nearly the entire coast redwood range, particularly in places which presently have fewer fog hours per day.
- Sea and land skin temperatures are increasing, which can have opposite effects on fog. A "climate haven" of stable temperatures exists along much of the California coast.
- Remote sensing is useful for spatially understanding fog frequency, filling the gap that in situ stations leave, but further research into climate change's fog impact is needed.

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